

QuestionsLab1.1.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



[Accede al documento original](#)

 **Gemini 3**

Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research 



Fuentes 

 Archivos

Estudia con Gemini

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

Expert

2 DÍAS
CONSECUTIVOS**86€**

Long expert

5 DÍAS
DE LIBRE USO**185€**Más info.:
elsubidon.es

If I want to execute the OpenMP version of Pi in boada with 8 threads, I can do it in two ways: interactively or via the execution queue. Which of the following affirmations is correct?

- ☐ The interactive execution on a node of boada will only be slower than the queued execution when the number of users using the node in the machine at that moment is large.
- ☐ I should not execute interactively Pi using 8 threads. But if I do, the system will report an error and execute only with two threads.
- ☐ The elapsed time of a program when executed through the execution queue will depend on the number of programs waiting in the queue at that moment. However, the CPU time will not depend on that, which is why we should only look at the CPU time (user plus system, as reported by the `/usr/bin/time` command).
- ☒ The rest of the options are false.

Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

🚩 Marca la
pregunta

Based on the information reported by lstopo for one of the nodes in the execution partition, which one of the following affirmations is correct:

- ☐ The architectural plot reported by the lstopo command shows two packages (sockets) each with half of the memory of the node (63 GB). All the cores in the package share access to this memory. However, as shown in the plot generated, one core in a package cannot access the memory that is available in the other package.
- ☐ The number of "threads per core" that is reported allows me to use additional OpenMP threads per core to execute my program in parallel. This always translates to better performance in its parallel execution because it allows better use of the hardware resources inside the core.
- ☐ All nodes in the execution partition of Boada are identical in terms of number of cores and memory hierarchy available and min/max frequency cores can execute our programs. At any time, one node in the execution partition can be executing different jobs, always ensuring that cores inside the node are not shared by the different jobs.
- ☐ All other answers are false.



WUOLAH

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00

🚩 Marca la
pregunta

When looking at the strong scalability plot of the Pi program, obtained after the execution of the submit-strong-omp.sh script, modified to be executed with up to 64 threads, we observe that (choose just one)

- ☐ There is a performance problem when starting to use more than one thread per core since the plot goes down abruptly when going from 32 (number of physical cores) to 33 threads.
- ☐ The speed-up and execution times for 32 and 64 threads are approximately the same because this program does not benefit from having more than one thread per core.
- ☐ The weak scalability plot shows a speedup smaller than 1 (compared to the execution time lasted by one thread) in all cases with a slightly negative slope. This is because there is some overhead per thread that impacts the useful work that each thread can do to solve the problem. Ideally, the execution time of the application should remain constant.
- ☐ All other answers are true



WUOLAH